

- Note:-** (1) Even if the casualty is breathing but the breathing is not normal, it is wise to start artificial respiration.
(2) Do not begin heart compression until you are sure that the heart has stopped beating.

5.8 External Heart Compression or External Cardiac Massage:

- a. Place the casualty flat on his back on a hard surface and remove the cloths over the chest.
- b. Sit on the right side of the casualty
- c. Feel and mark the lower part of the sternum.
- d. Place the heel of your left hand on the marked point make sure that the palm and fingers are not in contact with chest.
- e. Place the heel of the right hand over the left hand.
- f. Push the sternum towards the spine. It can be pressed upto 1 to 1.5 inches.
- g. Adults should be given about 100 compressions per minute. For children from 2 to 10 years compressions with one hand heel will be enough, but compressions should be @ 100 times per minute. For infants below 2 years compressions with two fingers is good enough and applied at a rate of 100 times per minute.
- h. Press firmly but carefully, carelessness may cause injury to ribs.
If the treatment is effective
 - i. colour will become normal.
 - ii. Pupil will contract.
 - iii. Carotid pulse begins.

5.9 CPR (Cardio Pulmonary Resuscitation)

If heart and breathing both have failed give CPR - Cardio Pulmonary Resuscitation: Give 30 heart compressions of cardiac massage, then two inflations of artificial respirations and repeat the process.

1. SHOCK

Definition:

Shock is severe depression to vital functions of organs like brain, heart, lungs etc. due to less blood supply to the brain.

1.1 Causes:

- Wounds
- Fractures
- Burns & Scalds
- Snake bite
- Sunstroke
- Heart attack
- Dog bite
- Electrical shock etc.

1.2 Sign and symptoms of Shock:

1. Giddiness
2. Pale colour of face
3. Coldness
4. Cold clammy skin
5. Rapid and weak pulse
6. Nausea
7. Vomiting
8. Unconsciousness

1.3 General Treatment for Shock:

- a. Reassure (encourage) the casualty if he is conscious.
- b. Keep the casualty in supine position (face upward) with head lower than the body to supply more blood to brain. The head must be turned to one side to avoid tongue falling back and blocking the wind pipe.
- c. Cover with a blanket to keep victim warm. Do not use hot water bottles, massage or give alcoholic drinks.
- d. Give pain relievers and arrest bleeding, if necessary.
- e. Arrange medical aid.

2. WOUNDS AND HEMORRHAGES (BLEEDINGS)

Definition: Wound is caused due to breakage of skin / tissue.

2.1 Types of Wounds:

- Contused Wounds
 - Lacerated Wounds
 - Punctured Wounds
 - Incised Wounds.
- a. Contused wounds are caused by blunt instruments where there is no opening. It is treated by putting ice over the wound.
 - b. Lacerated wounds are caused by irregular edges of instruments like glass pieces, metal pieces, machine injuries, animal bites and are said to occur where the edges of the wound is irregular. Clean the wound with water and pick any floating foreign body. Cover the wound with a clean cloth or apply antiseptic cream.
 - c. Incised wounds are caused by sharp edged instruments like razor / knife where the edge of wound is in a straight line. Arrest the bleeding immediately if loss of blood is profuse.
 - d. Punctured wounds are caused by sharp edged instruments like needles, nails and gun-shot wounds where the wound is more deep than wide. The wound should be covered and packed with a pad.

2.2 Types of Bleedings (Hemorrhages):

- Artery Bleeding – Bright red in colour and flow in jets.
- Vein Bleedings – Dark red in colour and flow continuously.
- Capillary Bleedings – Red in colour and oozing from all parts of wound.

2.3 According to the place of wound bleeding are two types:

- External bleeding
- Internal bleeding

2.4 Danger of Wounds:

- It allows precious blood to escape from body.
- It permits harmful bacteria/virus or other injurious agents to enter into body.

2.5 Methods to arrest Bleeding:

- a. Direct Pressure Method: Whenever a person, has a bleeding wound which is free from any foreign body, direct pressure method is to be used for arresting the bleeding. Clean the surroundings of the wound. Put a dressing/pad and press the wound. Bleeding will cause.
- b. Indirect Pressure Method: It is applied by two ways :-
 - By using Tourniquet bandage,
 - By pressing Pressure Points

A Tourniquet bandage is used to stop bleeding for hands and legs only. It should be tied above the wound towards the heart in single bone area, and should be relaxed once in 15 minutes.
- c. Important Pressure Points:

For other parts pressure points to be pressed.

 1. Carotid pressure point on the neck on either side of voice box.

2. Sub-clavian pressure point on the inner end of collar bone.
 3. Bronchial pressure point on the inner side of upper arm.
 4. Femoral pressure point on the thigh bone.
- 2.6 **General Treatment (First Aid) to Wounds:**
- a. Place the victim in sitting/lying position and elevate the injured part if possible.
 - b. Expose the wound and clean the wound and surrounding area but do not disturb blood clot if already there.
 - c. Remove any foreign bodies which are floating.
 - d. Arrest bleeding by applying pressure directly on the wound by putting a pad or by the indirect method as the situation demands.
 - e. Apply antiseptic cream, dressing and bandage.
 - f. Immobilize the part where possible.
 - g. Give pain relievers and treat for shock.
 - h. Arrange medical aid.
- 2.7 **Nose bleeding Treatment (Hemorrhage):**
- a. Advise the victim to breath through the mouth.
 - b. Place the victim near a window or against a current of air in the sitting position with the head slightly bent forward.
 - c. Pinch the junction of the nose just below the hard part.
 - d. If available put ice piece over the nose or a wet cloth.
 - e. Warn him not to blow the nose.
 - f. Do not block the nostrils.
 - g. Arrange medical aid.
- 2.8 **Ear bleeding Treatment:**
- a. Place the victim on his back.
 - b. Do not block the ear passages
 - c. If one ear is bleeding turn the head to the same side from which blood is coming out, and see that the affected ear is down.
 - d. If both ears are bleeding, keep face upward and head little bit low for free? drainage of blood and raise the legs.
 - e. Do not block the ear.
 - f. Arrange medical aid.
- 2.9 **Internal Bleeding:**
- a) Symptoms of internal bleeding
 - Giddiness.
 - Skin becomes pale, cold and clammy.
 - Pulse gets rapid but very weak.
 - Sweating, thirsty, feels vomiting sensation.
 - Becomes unconscious.
 - b) Treatment for internal bleeding:
 1. If the person is unconscious, check air-way, breathing and circulation of blood. If any failure is noticed, restore them.
 2. Lay him on his back and raise the legs by using pillow to enable the blood supply to reach the brain.
 3. If he is conscious lay him on his back and raise the legs by using pillow to enable the blood supply to the brain.
 4. Shift him to hospital as early as possible.
- 2.10 **Head Injury:**
- As a result of head injury, blood and brain fluid may flow out of the nose, ear or mouth.
- a) **Symptoms:**
- Giddiness.
 - Skin becomes pale, cold and clammy.

- Pulse gets rapid but very weak.
- Sweating, thirsty, feels vomiting sensation.
- Become unconscious.

b) Treatment:

- Ask the person not to blow his nose.
- Do not pack ear or nose.
- Lay the person on the affected side.
- Ensure tongue does not fall back.
- If any ear bleeding noticed, do not block the ear passage.

3. BURNS AND SCALDS

Definition: Burn is an injury caused by

- Dry heat – such as fire / flame
- Friction – touching speedy moving objects
- Corrosive (burning nature) chemicals like acids / alkalis
- Touching an object which was charged with high tension electric current.

Scald is an injury caused by moist heat such as hot water, milk, oil, tar, steam etc.

3.1 Degrees of burns:

- | | | |
|------------|---|------------------------------------|
| 1st degree | - | Redness of skin, blister formation |
| 2nd degree | - | Internal tissue damage |
| 3rd degree | - | Complete charring of part. |

3.2 General treatment for burns and scalds:

- If a person's cloth catches fire do not allow him to run. Pour plenty of water or gently place him on ground and roll him slowly to put on the flames.
- Cool and clean the affected area with wet cloth / cotton or flood with water or dip the affected area into water, if it is possible.
- Remove bangles, rings, watches immediately, as otherwise it may not be possible to remove them later.
- Remove the burnt cloths by cutting those not sticking to the skin.
- Cover the area preferably with clean cloth but do not disturb blisters.
- Do not apply antiseptic cream for major burns
- If he is conscious give water with pinch of salt to make good lost salt and water. Weak tea with more sugar also may be given if he is not a diabetic patient.

4. FRACTURE, DISLOCATION, SPRAIN AND CRAMP

Definitions:

- Fracture is breakage, crack / bend of a bone.
- Dislocation is displacement of one or more bones from joint.
- Sprain is wrenching tearing of cartilage near a movable joint.
- Strain is over lapping of muscles at a particular place.
- Cramp is sudden painful involuntary contraction of voluntary muscles.

4.1 Causes of Fractures: Direct force, indirect force and muscular contraction.

4.2 Signs and Symptoms for Identification of Fracture:

- Pain
- Swelling
- Loss of power
- Deformity (change in shape or size).
- Tenderness (sever pain by gentle touch)

4.3 Types of Fractures:

- Simple Fracture means the broken ends of the bone do not come out by opening the skin and thus remain inside only.
- Compound Fracture means the broken ends of the bone come out by opening the skin and the fractured bone is in contact with outside air as a result of an injury.